

ILE DE GROIX, France

WMO: 072030

Lat: 47.652N

Lon: 3.502W

Elev: 177

StdP: 14.60

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	33.2	35.4	24.7	18.9	36.5	27.8	21.9	37.6	62.8	47.3	56.9	47.0	17.2	70	0.754

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	78.3	66.4	74.8	65.0	72.1	64.0	67.9	75.3	66.4	72.1	65.3	70.2	12.5	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	94.7	69.8	64.4	91.1	68.7	63.3	87.6	67.7	32.3	75.8	31.1	72.4	30.3	70.2	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 50.2	41.8	35.2	DB	30.3	85.5	3.6	4.8	27.7	88.9	25.6	91.7	23.6	94.4	21.0	97.8	(4)
(5)			WB	27.6	70.9	3.4	2.7	25.1	72.9	23.2	74.5	21.3	76.0	18.9	77.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.4	46.4	46.4	48.9	52.5	57.3	62.0	64.7	64.5	62.9	58.1	52.4	48.1	(6)	
	DBStd	8.00	4.94	4.45	4.19	4.00	4.05	4.32	4.15	3.13	3.55	3.78	4.29	5.28	(7)	
	HDD50	445	127	111	67	16	1	0	0	0	0	2	24	97	(8)	
	HDD65	3624	577	522	499	374	241	112	55	45	84	215	377	523	(9)	
	CDD50	2419	15	9	33	92	227	360	457	450	386	253	98	39	(10)	
	CDD65	123	0	0	0	0	3	22	47	30	20	1	0	0	(11)	
	CDH74	403	0	0	0	0	14	95	160	86	47	1	0	0	(12)	
	CDH80	74	0	0	0	0	2	23	30	15	4	0	0	0	(13)	
(14) Wind	WSAvg	16.9	19.2	19.3	17.7	15.9	15.4	15.1	15.4	14.7	14.7	16.3	19.3	19.8	(14)	
(15) Precipitation	PrecAvg	33.1	4.1	3.0	2.3	2.5	2.2	1.8	1.9	1.9	2.2	3.7	3.9	3.8	(15)	
	PrecMax	41.8	8.2	6.9	7.4	6.4	5.2	3.8	4.2	4.2	5.7	6.9	8.0	7.6	(16)	
	PrecMin	23.8	0.7	0.6	0.2	0.6	0.5	0.2	0.2	0.3	0.2	0.7	1.0	0.8	(17)	
	PrecStd	4.4	2.1	1.7	1.5	1.5	1.1	1.0	0.9	1.1	1.5	1.5	1.8	1.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.9	56.6	65.6	70.7	75.3	83.5	83.7	81.9	79.0	69.3	62.5	58.0	(19)	
		MCWB	53.9	52.2	53.5	59.1	63.3	68.4	68.1	68.4	65.6	63.6	58.7	55.1	(20)	
	2%	DB	54.6	53.9	59.3	65.2	70.2	75.9	78.1	75.8	74.4	66.3	60.6	56.7	(21)	
		MCWB	53.0	51.0	52.1	56.2	61.1	65.2	66.6	65.6	64.3	61.5	58.2	54.3	(22)	
	5%	DB	53.6	52.8	55.8	61.2	66.6	72.3	74.6	71.8	71.0	64.4	59.3	55.8	(23)	
		MCWB	51.9	50.5	50.9	54.4	59.7	63.8	65.6	64.5	63.1	60.4	56.8	53.6	(24)	
	10%	DB	52.7	51.9	53.8	58.4	63.6	68.9	71.5	69.5	68.5	63.0	58.1	54.8	(25)	
		MCWB	50.9	50.0	50.4	53.0	58.1	62.3	64.5	63.2	62.1	59.6	55.4	52.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.7	53.3	55.3	60.4	64.8	70.3	70.0	69.3	67.9	64.6	60.5	56.4	(27)	
		MCDB	55.1	54.9	62.4	67.8	72.3	79.3	80.2	78.2	75.3	68.0	61.4	57.1	(28)	
	2%	WB	53.6	52.1	53.4	57.3	62.3	66.7	67.8	67.1	66.1	62.9	58.7	55.3	(29)	
		MCDB	54.3	53.1	56.6	62.9	67.9	73.9	75.2	72.9	70.9	65.1	59.9	56.2	(30)	
	5%	WB	52.5	51.3	52.3	55.4	60.5	64.7	66.4	65.7	64.9	61.7	57.5	54.4	(31)	
		MCDB	53.3	52.3	54.5	59.8	65.3	70.8	72.3	69.9	69.0	63.6	59.0	55.5	(32)	
	10%	WB	51.4	50.5	51.3	53.9	58.8	63.0	65.2	64.5	63.6	60.5	55.9	53.3	(33)	
		MCDB	52.4	51.7	53.1	57.3	62.8	67.9	70.3	68.1	67.1	62.3	57.8	54.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	7.2	8.8	10.0	10.0	10.4	10.1	10.0	10.7	8.1	6.4	6.0	(35)	
		MCDBR	5.1	7.6	12.2	14.9	15.1	17.0	16.9	14.4	14.8	9.7	6.6	5.0	(36)	
	5% WB	MCWBR	5.8	6.5	7.6	8.7	8.8	9.3	8.7	7.7	7.5	6.4	6.3	5.5	(37)	
		MCWBR	5.0	6.2	9.5	12.2	13.2	14.9	13.5	11.7	11.7	8.1	6.1	4.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.334	0.347	0.372	0.384	0.389	0.386	0.383	0.373	0.361	0.360	0.339	0.327	(40)	
		taud	2.425	2.407	2.352	2.336	2.368	2.402	2.411	2.458	2.473	2.465	2.445	2.402	(41)	
	Edn at Noon	Edn at Noon	234	257	267	275	278	278	278	276	269	249	230	220	(42)	
		Edh at Noon	22	27	33	37	37	36	36	32	29	25	21	20	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	334	626	1004	1505	1783	1958	1871	1628	1296	738	423	291	(44)	
	RadStd	RadStd	32	62	101	147	120	179	85	108	89	61	36	32	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page